

Q. Develop a decision tree model to classify species in the *Iris dataset*. Evaluate the model using accuracy and a detailed classification report with precision, recall, and F1-score for each species. Use `plot_tree()` to visualize the decision tree.

Ans:

Model Accuracy Score: 0.9800 (98.00%)

CLASSIFICATION REPORT

Detailed Performance Metrics for Each Species:

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	37
Versicolor	0.96	0.98	0.97	51
Virginica	0.98	0.97	0.98	62
accuracy			0.98	150
macro avg	0.98	0.98	0.98	150
weighted avg	0.98	0.98	0.98	150

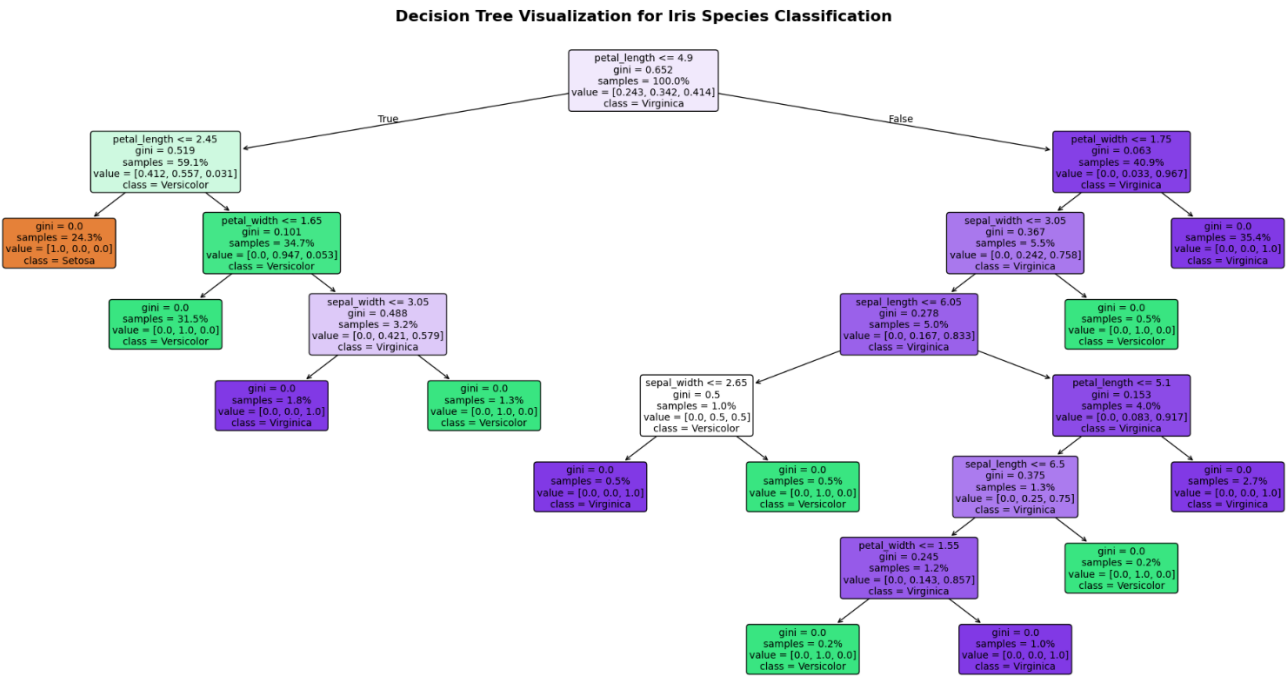


Fig: Decision Tree Visualization

Tree Depth: 7

Number of Leaves: 12